

PD 2.1A - Bimonthly Progress Report

Cancer Pharmagenomics

Introduction

The scope of this project focuses on the analysis of a Genomics of Drug Sensitivity in Cancer dataset from Kaggle that highlights thousands of lines of patient data. This information includes drug response and genomic profiles that allows for the investigation of relationships between genetic features and drug sensitivity. This project will use various data analysis techniques to uncover interesting relationships between the data.

Progress

Individual Preprocessing

Mohamad- How i started my preprocessing is by loading both datasets and i removed the rows with missing values under the cancer type(TCGA_DESC) column because this feature is required and can not be guess so any rows with missing values in these columns have to be removed. I then merged the two datasets using COSMIC_ID and DRUG_ID so matching cell line drug experiments align in one really big table. Finally, the selected categorical feature like cancer type and drug ID were encoded into numeric values using ordinal encoding, since machine learning models cannot work directly with text categories. The result is a clean, structured dataset ready for modeling.

Allan - I loaded three datasets into the file, GDSC_dataset.csv as the primary dataset , model_list_latest.csv, and genomic_features.csv. The first cleaning step was to drop AUC and Z_SCORE immediately, since both are derived directly from LN_IC50, the prediction target, and keeping them would have been bad for the modeling part. TCGA_DESC had 1,067 missing values, which were filled using the Cancer Type column, Tissue Descriptor 1 as a fallback, and individual cell line remaps, finishing with 31 clean labels and zero missing values. Tissue Descriptors 1 and 2 both had 9,366 missing values covering 39 cell lines with no annotation in the Kaggle file, which were filled using the Cell Model Passports file from Sanger by joining on COSMIC_ID rather than using statistical guessing. MSI Status, Growth Properties, and Screen Medium were all filled with Unknown as a safe placeholder first, then MSI and Growth Properties were upgraded with real Sanger values where available, leaving only rows where Passports also had no data as Unknown. The TARGET column had whitespace inconsistencies, causing duplicate categories, which were resolved by stripping whitespace, and CNA, Gene Expression, and Methylation availability flags were added as features. Finally, the genomic features file was pivoted from long format to wide format and merged into the main dataset on COSMIC_ID, bringing the final feature count to 750 columns across all 242,035 rows with no data lost.

Hiu - I loaded the merged GDSC dataset and selected LN_IC50 as the target variable. I dropped identifier columns like SANGER_MODEL_ID, COSMIC_ID, and DRUG_ID because they won't contribute to the prediction. To address the missing data, I created a scikit-learn preprocessing pipeline in which I applied StandardScaler() to the numeric features and one-hot encoding to the categorical features. Then, I split the data into training and test sets using an 80/20 split. After visualizing the categorical features, I found some class imbalance in categorical features, particularly the unclassified column in TCGA DISC features, which may require grouping rare categories in the next step.

Julie - I uploaded and read the GDSC dataset in my notebook then began preprocessing. Starting with exploring missing values where I found thousands. This discovery allowed me to find the relationship between variables. One of my discoveries was that many factors were dependent on cancer type and if cancer type was null many of the other lines were also null. Through additional findings I eliminated all missing values from the dataset because biological data is not accurate to average or assumed. Once my data appeared clean I began graphing my numerical features for additional exploratory data analysis. I plan to continue sorting through features and uncovering new relationships between variables in our data.

Em- After loading the main GDSC dataset and analyzing the shape of the data, including missing values, I removed rows with the missing/unclassified cancer type values. The data types are a mix of int, float, and object. We know that we want to include specific details about the biological and pharmacological profile of the cell line, and drop irrelevant data. We also need to scale numerical features and improve/optimize the encoding of object type data- using one-hot encoding. Then do exploratory correlation analysis.

Grace- I loaded the GDSC_dataset and the Cell_Lines_Details into my code. After I did this I was able to test for missing values and found a pattern between the missing values in both of the datasets. I started to develop code to find the correlation among the datasets. The first thing I did was make sure there was consistency among the COSMIC_ID column and standardized all of them. Then, based on the COSMIC_ID columns missing values, I merged those datapoints to avoid duplications among the missing values. I created a missing indicator data frame and an issue I ran into throughout that process was the fact that the runtimes were really high and crashed. The merging of the missing values is what is causing the crash, so I am currently working on this for an end goal of a correlation between all missing values. After I determine the correlation, I plan on starting to analyze the data.

Data Preparation

The Genomics of Drug Sensitivity in Cancer (GDSC) datasets used for this project were obtained via Kaggle, and originated from the Sanger Institute. The datasets are tabular CSV files

and the target variable, LN_IC50, is continuous. The main dataset includes key features like LN_IC50 (the log of the drug concentration needed to treat the cancer), TCGA (description of the cancer type) labels, and the drug names. The dataset contained a significant proportion of missing values and label inconsistencies.

One approach to addressing these issues was to load the main dataset and remove rows with missing or UNCLASSIFIED cancer types as well as dropping identifier columns that carry no predictive value. Categorical features were encoded into numeric values using either label encoding or one-hot encoding. The StandardScaler method was applied to numerical features. It was agreed on to flag and drop the UNCLASSIFIED category in TCGA_DESC as it did not represent a real biological classification. An 80/20 train/test split was completed by a member of the group which produced a final dataset of approximately 195,000 rows ready for modelling. The second, finalized approach was to, instead of dropping rows and missing values, fill them using additional supplementary CSVs from the Sanger Institute. The missing tissue descriptors, growth properties, and MSI status rows were cross-filled using the Cell_Lines.csv, while using the Genomic_Features.csv. These were merged to recover missing values and add 735 new gene mutations and CNA columns as additional features for the model.

The LN_IC50 (primary variable) column was explored first to check its distribution and missing values. It had a wide range of about -8.75 to 13.82 , with a heavy tail, meaning there were many extreme sensitivity/resistance values, though it presented no missing data that needed to be fixed. Missing values were checked for in other columns, which led to the finding of columns with missing values. The remaining columns were then checked for missing values, and found that the “PUTATIVE_TARGET” column had the most notable issues. Additional columns from other parts of the dataset were identified and incorporated into the final preprocessed dataset.

Methods

Several regression algorithms were implemented within the code, these regression algorithms help to model the relationship between the cancer type, drug identity, drug target and the target variable which measures drug response (LN_IC50). The target variable is continuous, so this means that the modeling was formulated as a supervised regression problem. The models selected are both linear approaches and non-linear tree-based methods, this allows the comparison of the learning models within the data.

The linear models that were selected include linear regression, ridge regression, and elasticnet. These models were used as the baseline models because they help interpret coefficients and they also are mostly used when working with high dimensional feature spaces that are created in categorical encoding features. Ridge regression includes the L2 regularization, which constrains coefficient size and it helps stabilize predictions when many correlated predictors are present. ElasticNet combines L1 and L2 regularization, which allows the model to reduce noisy predictors while still keeping the useful signal. These regularization parameters

help control the model to evaluate whether simple linear relationships between the predictors and drug response were good enough while still controlling the model's complexity.

Additionally the non-linear relationships and interactions between the features were also included, these were captured by using tree based ensemble models like HistGradientBoosting Regressor, LightGBM and CatBoost. These models work well on structured tabular datasets and can automatically model complex features, this is something that the linear models are lacking. These methods build many decision trees that help with focusing on correcting the residual errors from the previous trees, this improves the predictivity across the successful boosting. Most of the models were trained using default parameters initially, this helped establish and compare baseline performance for the different preprocessing strategies. For the tree based models, low or moderate learning rates and limits on the tree sizes were used to help with reducing the overfitting, at the same time it maintains the model flexibility. A great example of this, is limits on tree depth and minimum samples sizes were used to prevent overly small splits in the data. The parameter choices push generalization by preventing overly complex trees that have the ability to memorize the training data.

All together these models allow the group to evaluate how different modeling and linear relationships versus nonlinear ensemble learning affect the ability to predict drug sensitivity from the categorical biological and pharmacological features. The comparison between the different models also provides insight into whether predictive performance was primarily driven by simple additive effects or by more complex interactions that are captured by boosting based algorithms.

Performance

The metrics used to evaluate the models were RMSE/Root Mean Squared Error, MAE/Mean Absolute Error, and R-squared. These three were chosen because LN_IC50 is a continuous target, so metrics are needed that measure both the magnitude of prediction error and the model's ability to explain the variance in the data. RMSE penalizes larger errors more heavily, which matters since extreme sensitivity or resistance values are clinically meaningful for the prediction. MAE gives a more interpretable average error in the same units as LN_IC50, and R-squared tells how much of the variation in drug response the model is actually capturing. For the initial modeling efforts, R-squared and RMSE evaluations were used to select the model. Since the group has focused on two different versions of Preprocessing, to fill the missing row values or drop them, the models are evaluated on both under default parameters.

For the dropping approach, Linear Regression and Ridge perform well, with an RMSE of 1.3914 and an R-squared of 0.7498, while LightGBM achieves an R-squared of 0.7221 and CatBoost follows closely with an R-squared of 0.7177. But the issue with these positive results from this step is that the dataset provided to these models would become biased, reducing confidence in their accuracy.

For the filling approach, the linear models struggle significantly with Linear Regression, having an r-squared of - 8.35 and Ridge - 5.73, showing they cannot handle the added complexity and noise from the filled and merged data. ElasticNet performs much better as a baseline with an RMSE of 2.0368 and an R-squared of 0.4581, showing it can at least capture some meaningful signal. The stronger model performs well here, with LightGBM reaching an RMSE of 1.3265 and an r-squared of 0.7702, which is the best overall result across both approaches, and also CatBoost comes in at an R-squared of 0.7105. These performances of this approach show a more biased approach and more assurances on the accuracy of these models, but it has to go through multiple obstacles to get to this point

Plan

Week 1:

- Determine the core feature set for the baseline models, such as encoded TCGA labels, key drug features, and a selected subset of genomic features.
- Conduct baseline regression experiments. Run controlled experiments that compare ElasticNet, linear regression, and ridge regression under 80/20 and 75/25 train-test splits, evaluate the results using RMSE, MAE, and R-squared.

Week 2:

- Implement and evaluate more advanced models, such as Random Forest, and XGBoost, and evaluate the metrics for comparison.
- Summarize the performance of all tested models (baseline and advanced) in a clear table that includes RMSE, MAE, and R-squared, and notes on the preprocessing strategy and feature sets.
- Start drafting the narrative that connects EDA findings, preprocessing decisions, and modeling results back to the core questions: Can we predict how sensitive a cancer cell line will be to a drug, given its biological and pharmacological profile?

Remaining Questions

Remaining questions at this phase of our analysis include diving deeper into connections between different variables and what these connections mean. Additionally, would additional hyperparameters improve the models performance?

Expected Results

In the next two weeks, the group is expected to complete that data preprocessing and decide on a model they are going to use to execute the rest of the project. Currently, the best method is unknown and will be tested. A promising model, ElasticNet, is being used to perform a regularized regression model to linearly apply the Lasso/Ridge concept with the dataset. This method is seeming the most efficient at the time and will be used to determine the expected

results. An ideal output would be an R^2 of around 0.3-0.5 indicating a meaningful signal from the model before ElasticNet outputs the best numbers out of the other models being tested so far, with an RMSE of 2.0368, MAE of 1.6327, and R^2 of 0.4581. The team will analyze the data using graphs of different parameters such as, LN_IC50, DRUG_NAME, TARGET_PATHWAY, to ultimately determine the genetic codes that lead to worse adverse side effects. After these kinks are figured out, since the model uses the default parameters, they will be fine-tuned in the next couple of weeks for better accuracy.